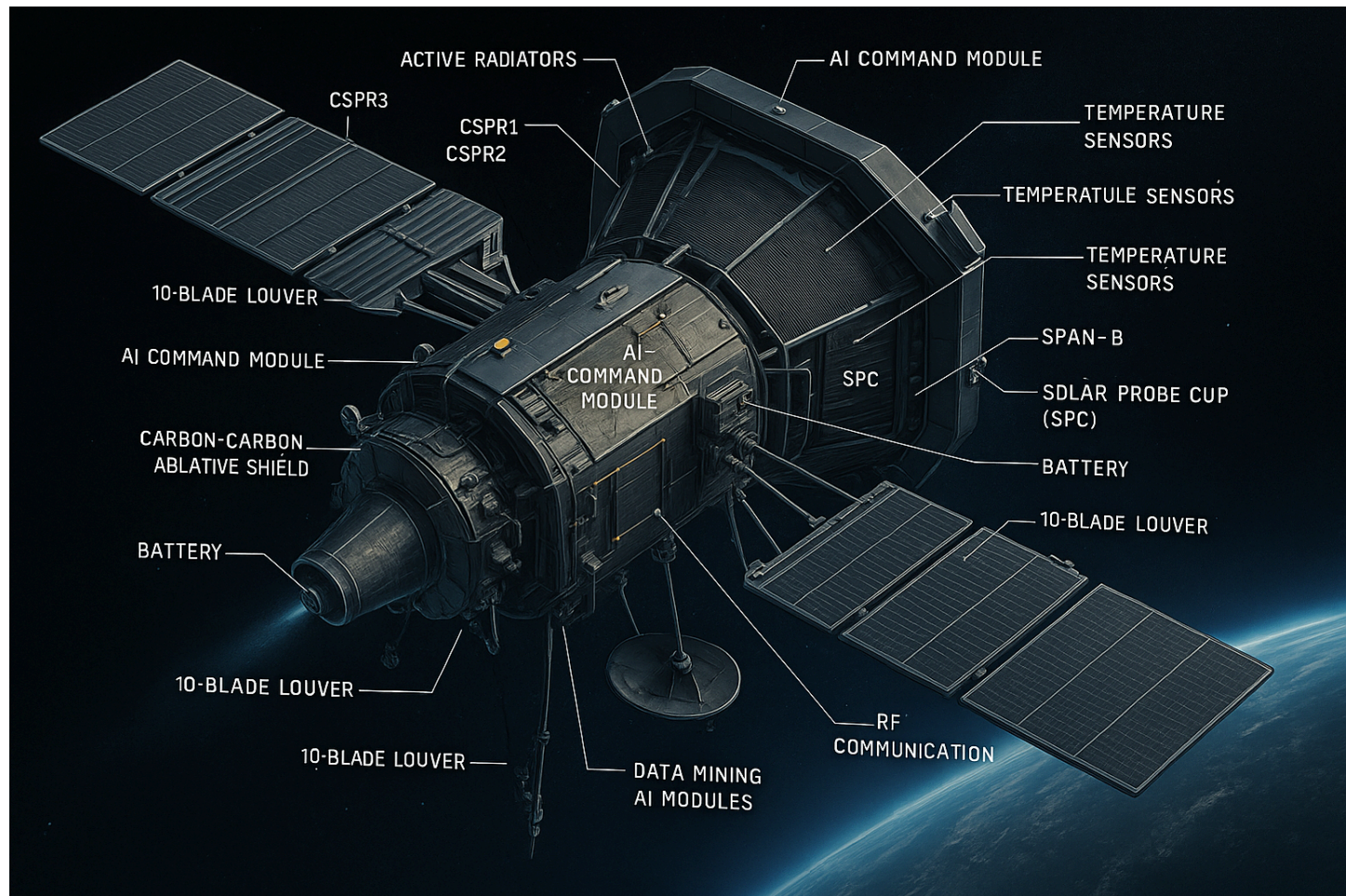


Combustion systems & fuel, composites, mechanical designs



VASIMIR Propulsion

High density ($>1019/m^3$), steady-state helicon discharges in hydrogen, helium, deuterium and argon are now routine and, as shown in Figures 3(a) and 3(b), already show a high velocity exhaust downstream of the magnetic throat. Flow velocities of 15 Kml sec, measured with a reciprocating Mach probe, are quite common and indicate a respectably high specific impulse even before ICRH power is added.

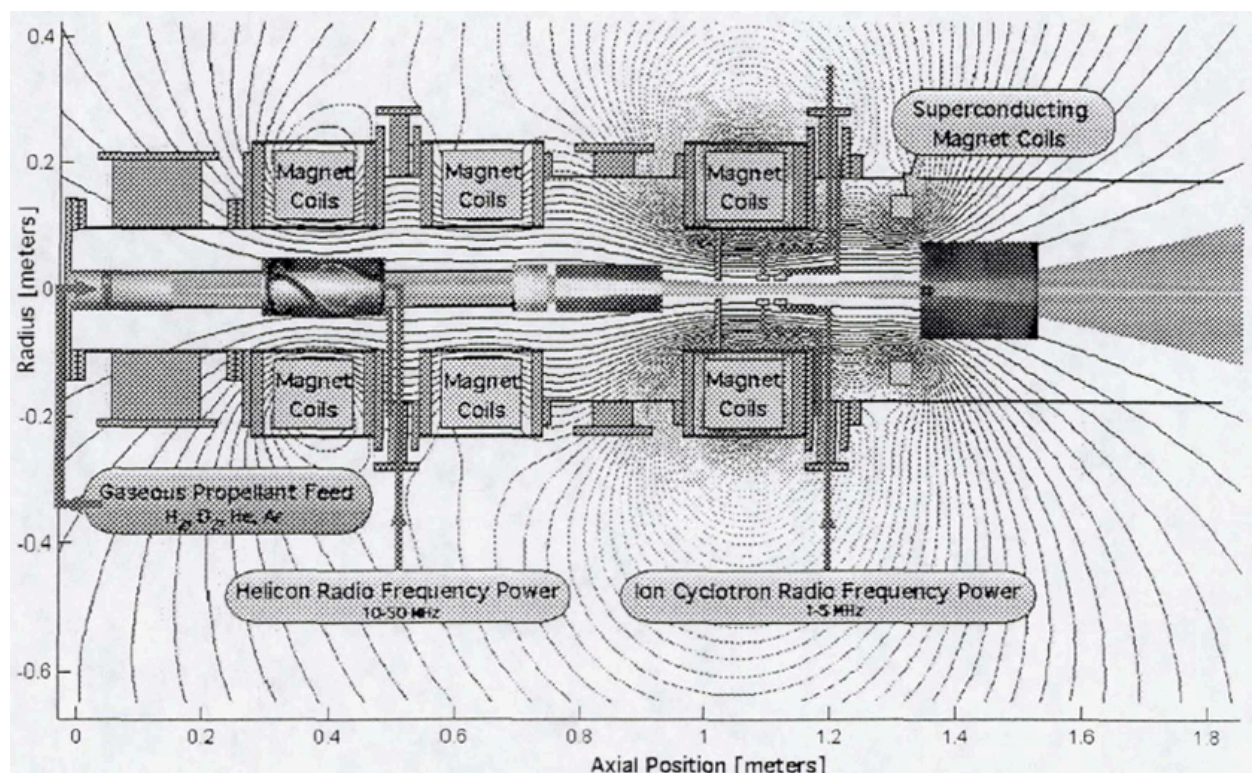
The functional design of a cold gas propulsion system. The formation flying of two satellites implies the following; the initial relative position and velocity is maintained while the orbital altitude of the satellites may degrade. The decay of the satellites is influenced by atmospheric drag and by the geo-potential field, the Earth's gravity field. Due to differences in position, area and mass of the satellites they drift apart and for this the propulsion system is adjusting [1]. Depending on the distance the satellites are allowed to drift apart, the frequency and accuracy of the corrections are obtained. These parameters can be translated into a minimum impulse bit. For formation flying the impulse bit is simply defined by: $I_b = \Delta$

JET ENGINE COMPONENTS

1. Diffuser - Slows down airflow and increases pressure.
2. Compressor - Increases air pressure via energy transfer from turbine.
3. Combustion Chamber - Location of air-fuel combustion.
4. Turbine - Converts thermal energy to mechanical energy.
5. Nozzle - Accelerates exhaust gases for thrust.

SPACECRAFT COMPONENTS

1. Jet Engines - Provide thrust.
2. Fuel System - Stores/supplies fuel.
3. Control System - Regulates engines/operations.



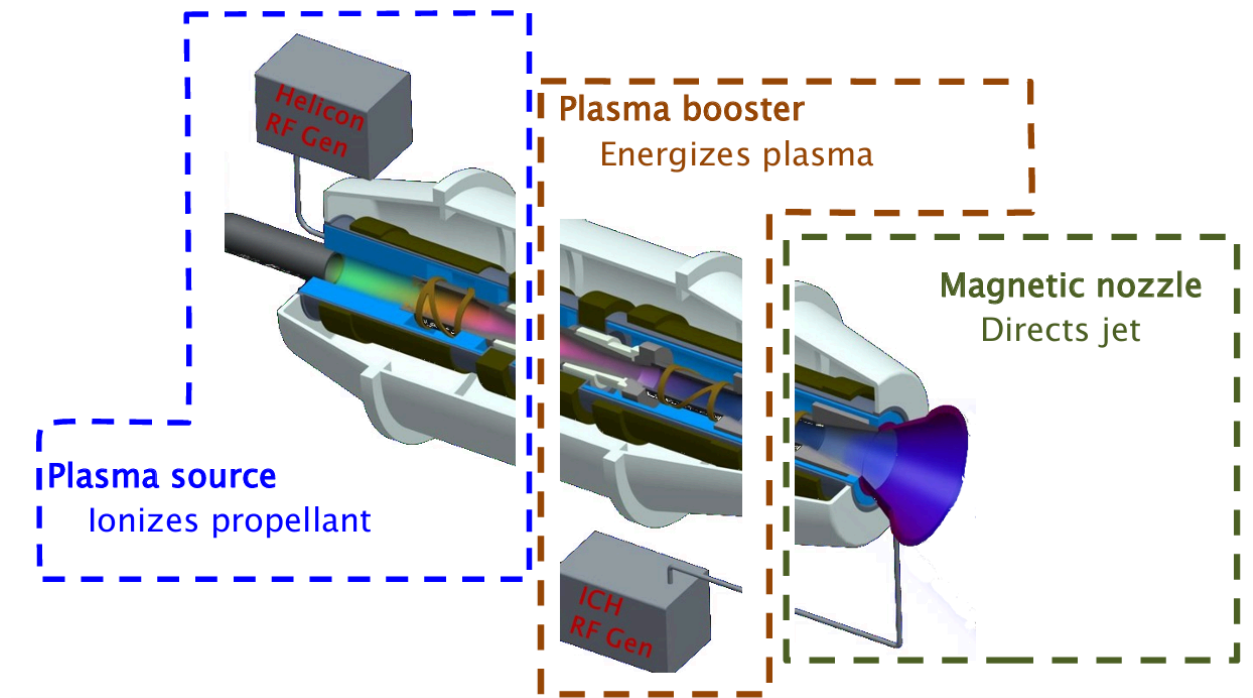
In the International Space Station, an experimental VASIMR operating at 25kW could provide sufficient thrust to neutralize the atmospheric drag on the facility. The system could operate on deuterium or hydrogen. VASIMR technology enables very fast human planetary transits to Mars and beyond. The development of the VASIMR and other advanced plasma rockets addresses a fundamental need in space transportation. Such an investment in technology will also benefit the commercial satellite industry, by enabling much higher payloads and mission flexibility. While a relative newcomer to the family of advanced rockets, VASIMR has evolved rapidly and its research team is now demonstrating important results in several areas. Such advances place it in a competitive realm with the more mature technologies of electrostatic plasma accelerators.

Receives power from solar or nuclear source: Directed plasma jet & generates thrust

Propellant options: H, He,
Ne, Ar, Kr, Xe...

Heated ions enter an adiabatic “nozzle” in the static magnetic field

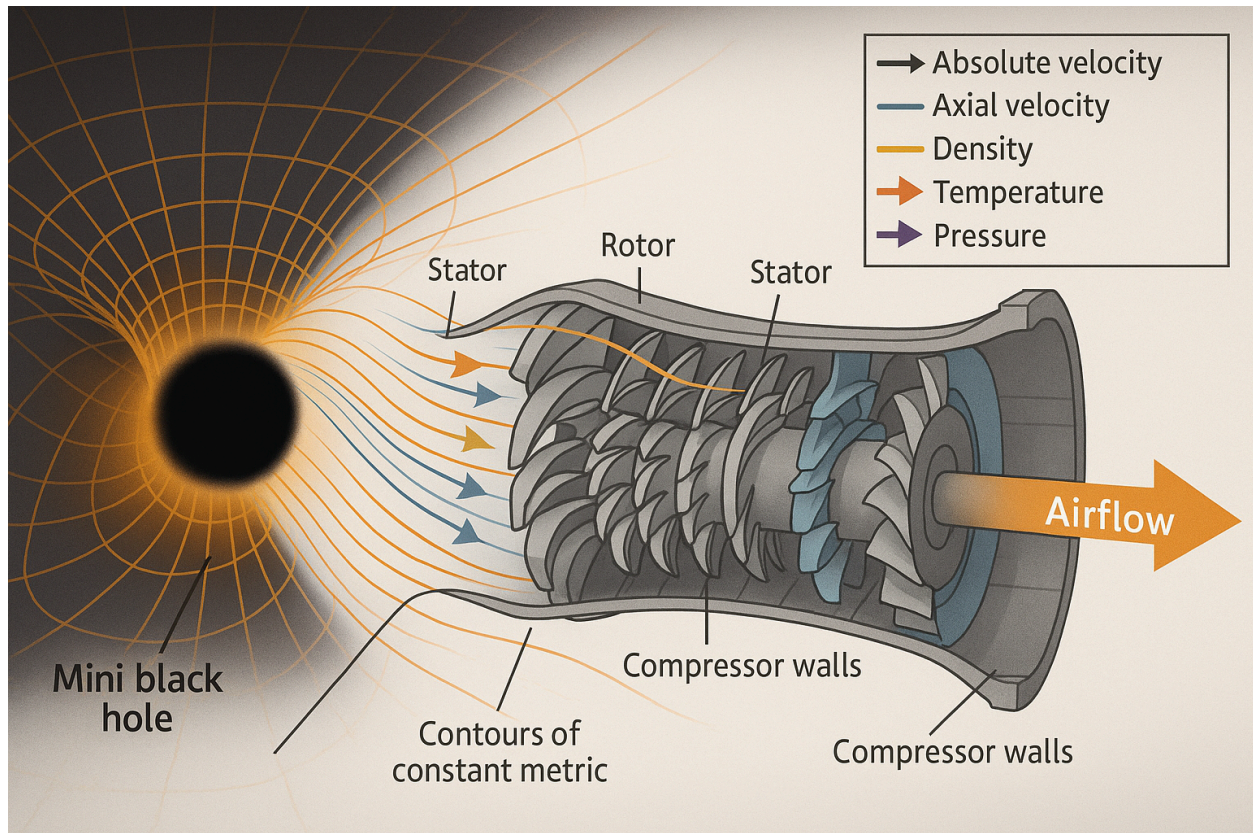
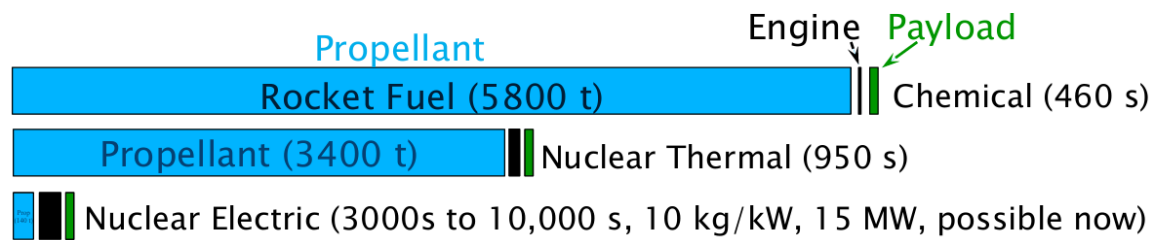
- Directs ~90% of the ion energy into an axial plasma jet for thrust



Uses RF waves to couple & power; VASIMR technology provides electric propulsion (EP) with the power density needed for sustainable human exploration

- The technology scales naturally to power levels for robotic and human exploration (> 40kW solar at first, scalable to multi-Megawatts)
- Specific impulse from 2000 s to 5000 s
- Total dry mass is approximately 2600 kg for 1 MW
- Adaptable to cover a broad range of mission

Requirements: Scalable to multi MW power levels High power density (6 MW/m²) Solar electric (near term and cargo) Nuclear electric (future)



Radiator mass reduction for space drives the design to high temperatures (Stefan Boltzmann law) The High thermodynamic efficiency requires large ΔT Materials limit the maximum Temperature. The Surface density can approach 1 kg/m² and specific mass below 0.2 kg/kW at 300 C

Steady-state RF PPU– Compact

Highly efficient ,Operated with both plasma stages TRL 5-6 (individual boards in vacuum)

Power rated: 48 kW

Efficiency: 91% (expect to increase to 95%)

Size: Weight:

40 cm x 40 cm x 120 cm

40.1 kg; $\alpha = 0.9$ kg/kW

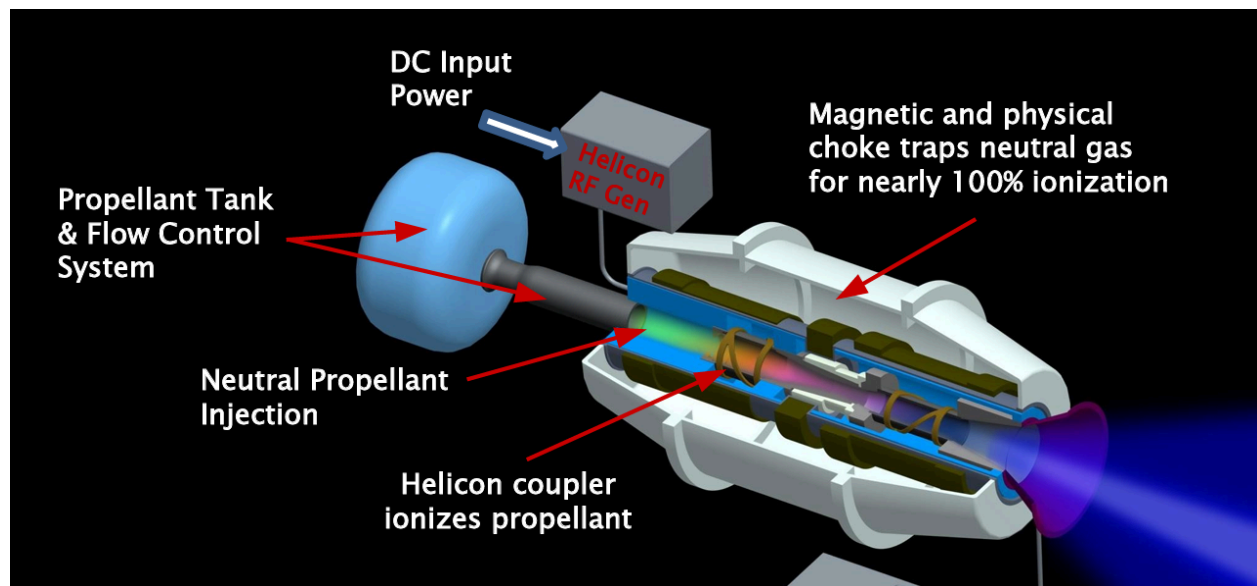
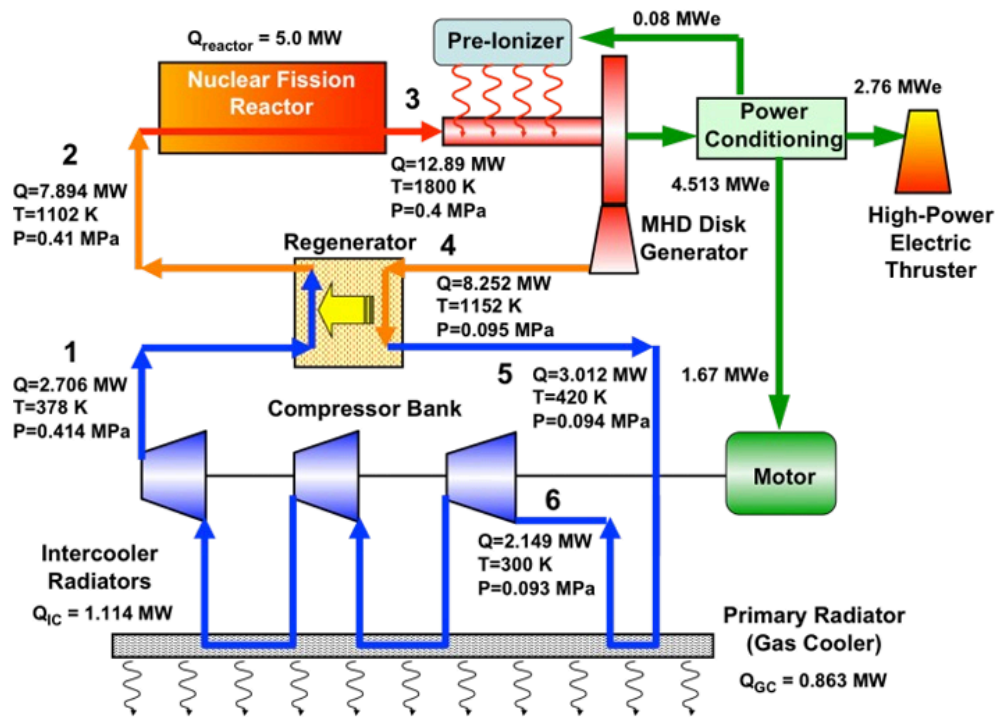
ICH

Power rated: 180 kW (24 hr ss burn in)

Efficiency:

98% now Size: Weight:

40 cm x 40 cm x 120 cm 87.1 kg; $\alpha = 0.5 \text{ kg/kW}$



Very high power density ($\sim 6 \text{ MW/m}^2$) compared with DC discharges by coupling RF power through ceramic windows to excite naturally occurring plasma waves

No DC bias is applied, no cathodes or anodes are needed. Plasma facing materials are well insulated by strong magnetic fields with field strength of a few tesla

Specific impulse can be varied at constant power by adjusting the balance between the plasma source and the ion cyclotron heating (ICH) booster sections

Wide range of operating power supported without loss of efficiency. Very efficient conversion of input power to jet power (75% thruster efficiency, >65% system efficiency) allows more thrust and less propellant usage than Hall thrusters for a given solar electric power plant

Wide range of propellant choices including xenon, krypton, argon, and neon. May also be adaptable to solids or liquids (ammonia, iodine, potassium, rubidium, etc.) if condensates do not pose a risk to the spacecraft.

Subsystem VASIMR® Hall

Gas Injection

- Single port injection
- Simple flow control
- Separate injection for anode and cathode
- Complex feedback required for flow control

Magnets

- Superconducting for high field with minimal power
- Well isolated from plasma
- Alignment with flow inherently provides surface protection
- Conventional, limits field
- Close proximity to plasma
- Alignment perpendicular to flow requires compromise for surface protection

Thermal Management

- Low-temperature fluid loop for power and avionics
- High-temperature fluid loop to manage thruster heat rejection
- Low-temperature fluid loop for power and avionics
- Very high-temperature to reject thruster waste heat by direct radiation

Called the “ion cyclotron” wave Ion Cyclotron Heating (ICH) uses resonance between wave and ion gyro-motion Resonance near the aft end of the booster

Magnetic field fully shields ceramic parts

Acceleration from ICH enhances the source

PROPULSION MATERIALS

1. Fuel & Oxidizers - Generate thrust.
2. Plasma Propulsion - Plasma-based systems.

BLACK HOLE RESEARCH

1. Spacetime Symmetries
2. Plasma Tensor Converter - Analyzes plasma-black hole interactions.

THERMAL CONTROL SYSTEM (SACS)

- Heat dissipation: 6400 W at 9.86 Rs
- Liquid Cooling Loop (Redundant Pump/Electronics)
- Temp Ranges:
 - Operative: +20°C to +150°C
 - Survival: +10°C to +190°C (wet), -80°C to -130°C (dry)
- Activation:
 - Prelaunch: Accumulator heated to 40–50°C
 - Radiators 2 & 3 active at 0.89 AU (Day 41)

Key Components:

- Accumulator (non-redundant)
- Pump Electronics (block redundant)
- Platens, Radiators, Valves
- Thermal Simulation & Testing (C1/C2 configs)

CALCULATIONS: COMPRESSOR STAGES

Centrifugal Compressor:

INLET:

$C1 = 150 \text{ m/s}$; $W1 = 250 \text{ m/s}$; $U1 = 200 \text{ m/s}$

OUTLET:

$C2 = 390 \text{ m/s}$; $W2 = 150 \text{ m/s}$; $U2 = 350 \text{ m/s}$

$\beta_2 = \text{Flow angle}$; $\beta_2' = \beta_2 + 5^\circ$

(a) Impeller Type:

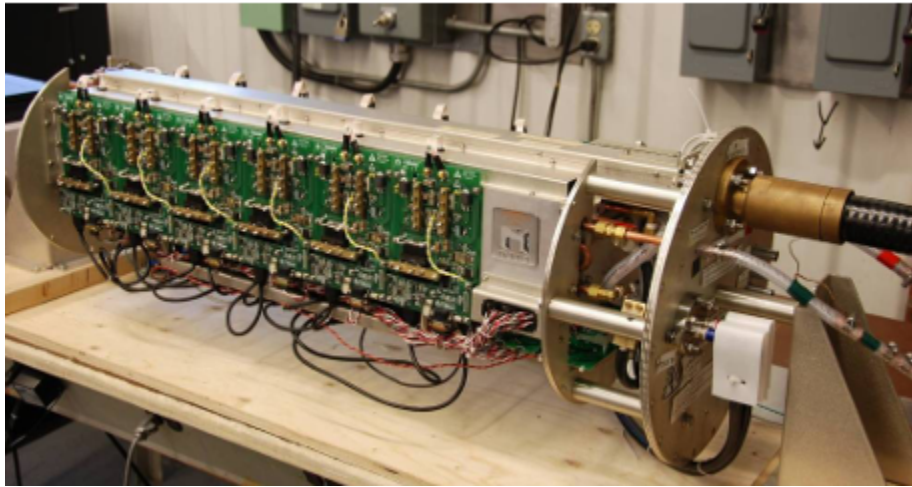
Based on the velocity triangle, calculate β_2 .

If $\beta_2 < 90^\circ \rightarrow$ Forward-curved

$\beta_2 = 90^\circ \rightarrow$ Radial

$\beta_2 > 90^\circ \rightarrow$ Backward-curved

Plasma propulsion systems (space & cars)



Notes: Experiments from 2009-2012 used a low temperature superconducting magnet 6 K

No coupling to plasma operation...Negligible current maintenance for steady magnetic field
High-temperature superconducting (HTS) magnet coil prototype developed and tested in 2007 (Creare, Tai Yang) Preliminary design of proto-flight coils and manufacturing test with 2nd generation HTS (SuperPower Inc.) in 2008 Performance improving with Moore's law Sunpower GT cryocooler baselined in 2013 preliminary design (200,000+ hours mean time to failure)

Notes: "Low temperature" system is pretty conventional ◦ Avionics, controllers, and cryocoolers with max temp ~35 °C ◦ RF generators with CSi modules can tolerate max temp up to ~85 °C to reduce radiator size "High temperature" system for rocket core ◦ Heat flux measured during VX-200 operation ◦ Rocket core has Integrated Cooling and Electrical (ICE) jackets ◦ Combination cooling function with the RF power couplers ◦ As high as 280 °C An infrared camera and thermocouples measured the temperature evolution inside the VX-200 plasma source during operation with no cooling. Transient analysis was used to identify heat flux and hot spots.

VASIMR® technology provides electric propulsion (EP) with the power density needed for sustainable robotic space operations and human exploration [Reference 2,4]

◦ The technology scales naturally to power levels supported by the National Research Council for robotic and human exploration (> 40kW solar at first, scalable to multi-Megawatts)

VASIMR® technology differs from traditional Hall thrusters and ion engines in that it relies on natural wave propagation in magnetized plasma to reach high power density.

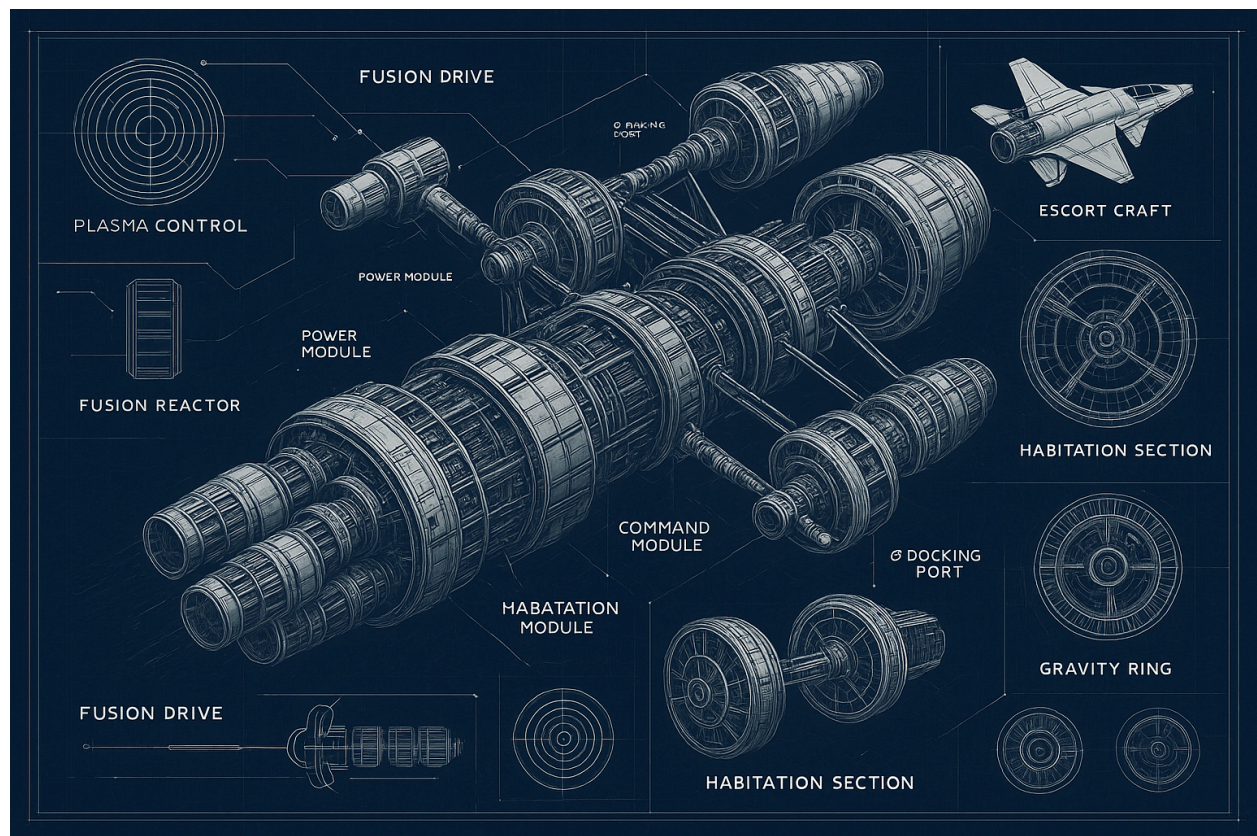
- Waves include whistler or high-harmonic fast waves (HHFW) and ion cyclotron waves
- Relaxes Child-Langmuir and Debye shielding constraints encountered in DC discharges

The technology has a long history of development (1979-present) with collaboration of academia (MIT, Princeton, UMD, UTexas, UMich, etc.), national laboratories (ORNL, LANL), international labs (Alfvén, ANU, Japan), NASA and private industry.

- All of the physics to support VASIMR as an advanced propulsion technology are well understood and have been demonstrated in the appropriate laboratory environment
- Experimental results have been presented and published in multiple peer reviewed journals over several years

- VASIMR® technology has been developed almost entirely with private funding (~\$30M) and has been patented, but a great deal of misinformation circulates about its status

VASIMR is now fully enabled by modern technology developments, particularly high-power solid state transistors and high-temperature superconducting (i.e. YBCO) tape.



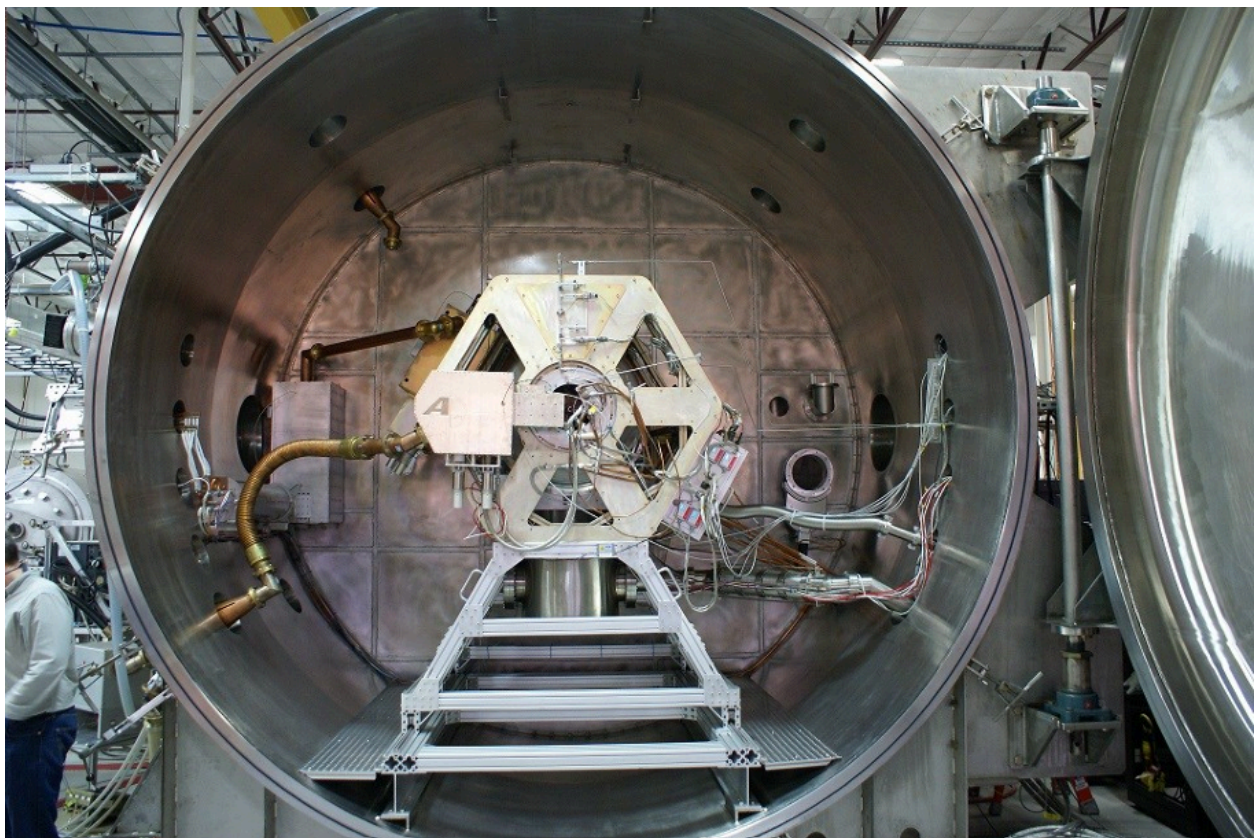
A. Ceramic Surface Temperature Measurements

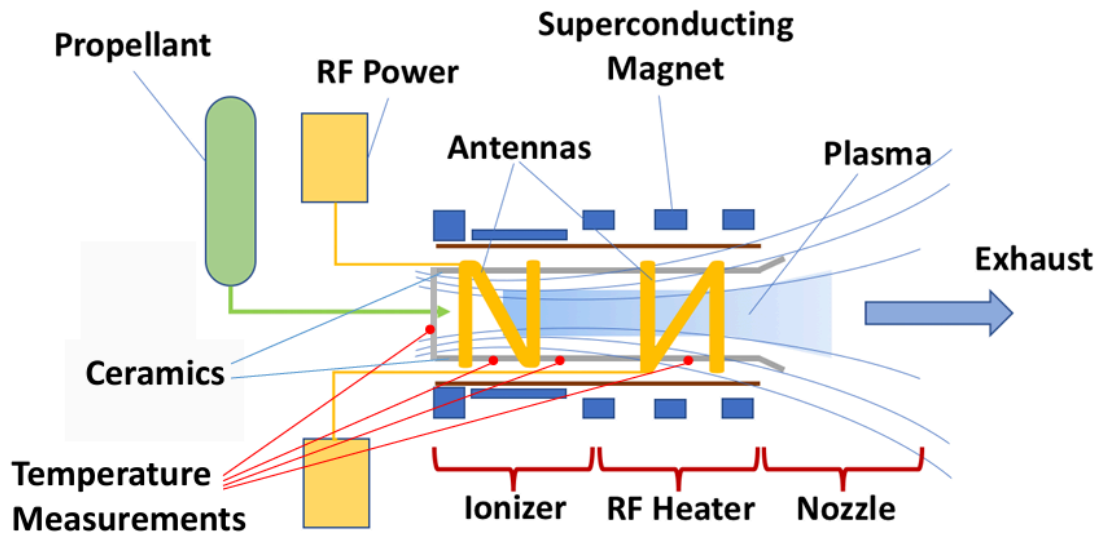
A conceptual schematic of a VASIMR® system is shown in Fig. 2. The 1st stage “Ionizer” is a helicon type plasma source. The 2nd stage “RF Heater” is an ion cyclotron heating device. Figure 2 also shows the temperature measurement locations of plasma-facing ceramics in the VX-200SS™ prototype. The ceramic windows confine neutral gas and allow RF power to be coupled through to the plasma, but they must tolerate the associated heat load. Ceramic temperature measurements are therefore important for monitoring the performance of the thermal design. Ad Astra has been using infrared pyrometers to measure plasma-facing component temperatures since 2019. Earlier efforts were thwarted by the challenges of the ambient environment within the rocket core.

B. Power Measurements

Each stage of a VASIMR thruster uses an independent radio frequency power system. Figure 3 shows a simplified schematic of the power circuit for the 2nd stage of the VX-200SS™ prototype. The circuit for The 1st stage is similar, except that the 1st stage power processing unit (PPU) is still outside the vacuum.

The locations where power can be measured in the circuit. In this paper, all presented power measurements were made with the DC current and voltage sensors located just to the left of the vacuum feedthrough. For the 1st stage circuit, the DC sensors are at the connection to the PPU.



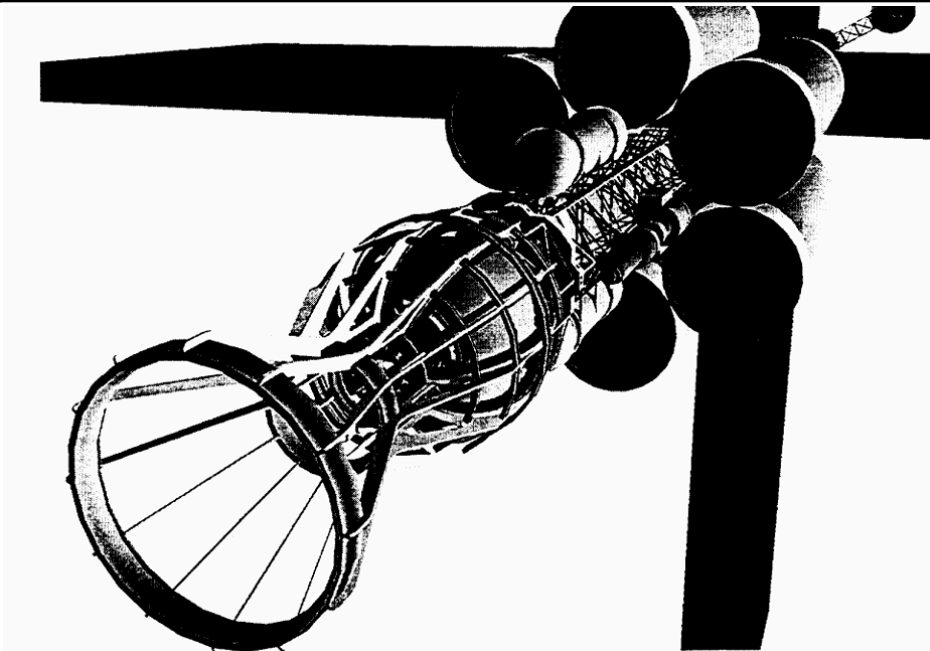


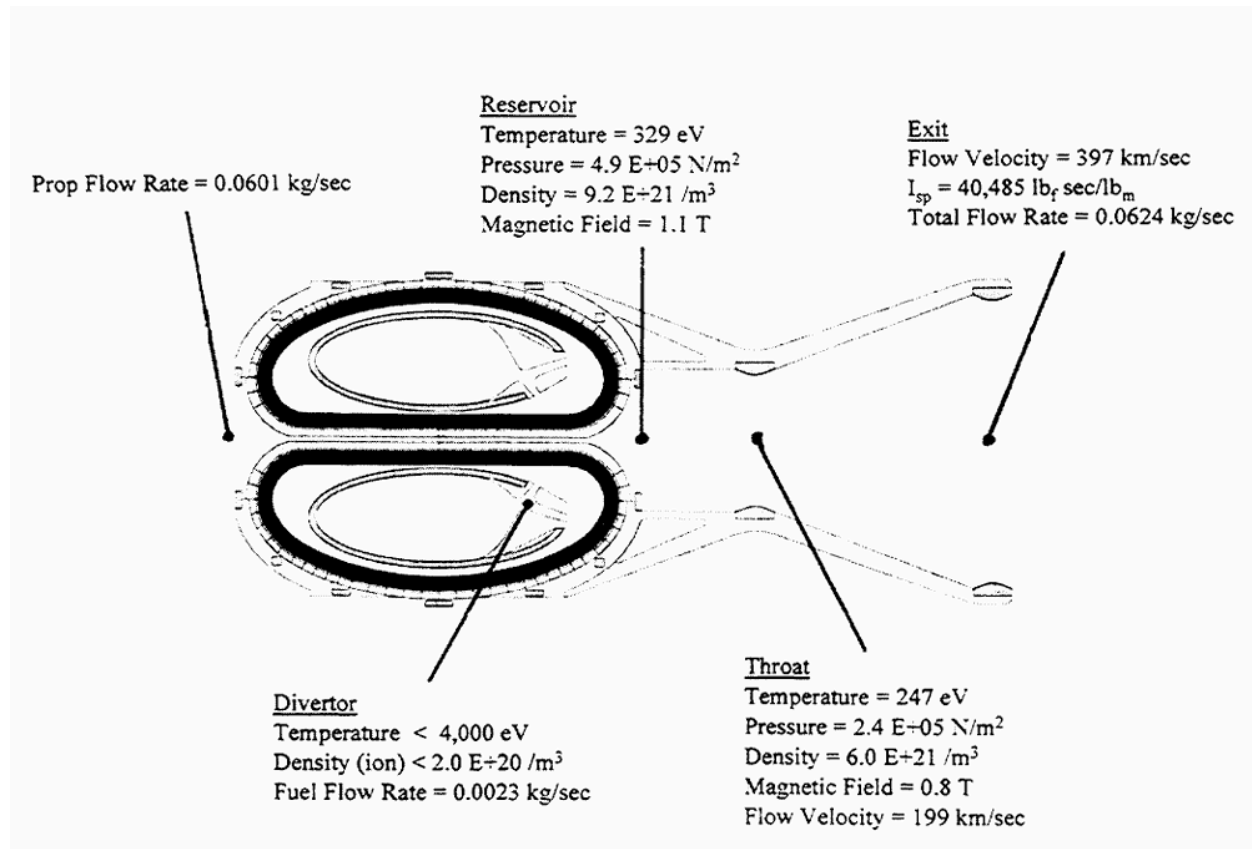
Propellant Sealing The plasma-facing ceramics of the VX-200SS™ are an assembly of several manufactured components. The ceramic-to-ceramic joints in this assembly must be sufficiently sealed to prevent leakage of neutral gas into the space between the rocket core and the magnet cryostat. If the pressure in this annular space becomes too high, the operational power of the VX-200SS™ will be limited by Paschen breakdown at the RF antennas. Designing seals for this application has been challenging; the seals must tolerate the temperature cycles of the ceramics, electromagnetic fields in the proximity of the radio frequency antennas, and the static field of the superconducting magnet. Figure 6 presents two measurements of partial pressure versus time during VX-200SS operation: one made in May 2021 and another in June. These mass spectra were measured by a residual gas analyzer in the upstream section of the vacuum chamber. During thruster operation, the upstream argon (40 u) pressure is due to a combination of argon exhaust leakage through the vacuum chamber partition wall and neutral propellant leakage from the rocket core seals. Since the vacuum pumping and partition wall seals are typically unchanged from one experimental campaign to the next, the evolution of ambient upstream argon pressure during thruster operation is a good indicator of the rocket core seal quality.

Fusion reactor system: It was designed to analyze small and large aspect ratio, tokamak fusion Inductive and non-inductive heating, driven and ignited operation, burning DT, DD, or D3He fuels.

By pursuing peaked temperature and number density profiles fusion within the core of a plasma, a relatively small producing region was established, satisfying Lawson and ignition criteria without necessitating large beta (13) throughout the plasma. The lower temperature and number density outer regions would contribute to a Volume-averaged 13 values within MHD stability limits.

Major radius (m)	2.48
Minor radius (m)	1.24
Aspect ratio	2.0
Elongation	3.0
Plasma volume (m ³)	225.8
Safety factor (edge)	2.50
Safety factor (axis)	2.08
Fuel ion density (10 ² /m ³)	5.0
Electron density (10 ² /m ³)	7.5
Plasma temperature (keV)	50
Volume averaged beta	0.318
Confinement and Plasma time (sec)	0.552
Average neutron wall load (MW/m ²)	1.03
Average radiation wall load (MW/m ²)	5.20
Ignition margin	1.235
Toroidal magnetic field (centerline) (T)	8.9
Maximum magnetic field (coil surface) (T)	52.1
Gain factor (Q)	73.1
Plasma current (MA)	66.22
Bootstrap current fraction	0.934
Wall reflectivity	0.98
Number density profile shape factor	1.0
Temperature profile shape factor	2.0





In space, a vehicle can accelerate itself by ejecting part of its mass. This acceleration mechanism is called thrust and a device that generates thrust is called a thruster. The nozzle is a key component of the thruster because it increases the efficiency of the system. In the way the nozzle is shaped, it increases the supersonic velocity of a gaseous exhaust. With the same amount of expelled mass more thrust is generated and thus efficiency is improved. There are several methods by which thrust can be generated, not all of them are suitable for miniaturization.

In a solid propellant system an array of small chambers, each having an igniter and a nozzle, is filled with a solid material. The solid propellant decomposes when enough heat is supplied. After ignition the hot gaseous products leave the chamber through the nozzle. In this way a predefined amount of impulse is generated and it is therefore also called digital propulsion.

The following mechanisms are based on a propellant - gas or liquid - that is stored at high pressure. Besides a pressurized fuel tank such a propulsion system can consist of several components, i.e. filters, valves, pressure regulators, tubing, heaters and thrusters. For health monitoring, pressure sensors, flow meters, and temperature sensors might be necessary. Depending on the type of propulsion, this can be quite a complex system. Three different configurations of propulsion systems are discussed in order of increasing complexity and efficiency and all utilizing a pressurized gas tank.

Cold gas propulsion Cold gas propulsion is the simplest configuration, requiring the least components. In a controlled manner gas is released from a tank by a valve and expelled through the thruster. Such a cold gas blow-down rocket engine typically consists of a gas tank, a pressure and temperature sensor, an on/off

valve and a thruster. If the tank contains a gas at a very high pressure a pressure regulator is required in between the tank and the valve to reduce the pressure to a convenient operation level. Grönland applied MST to build a cold gas thruster pod [23]. Another such system is presented by Köhler [24]. Although no MST was used, Gibbon and Wart presented a miniaturized propulsion system based on the vaporization of butane [25]. Butane is stored as a liquid propellant and is vaporized at a pressure of 3.8 bar at 40°C. The heated vapor is expelled through the thruster. Where Δv (delta-v) is the change in velocity and m the mass of the satellite. The impulse bit is also expressed as the force T , which is actually the thrust, times t_{min} ; the smallest time unit.

The total delta-v is then calculated by: mission duration $\Delta v = t \Delta v$ [m/s] (2.2) period between corrections:

Power Conversion	
Mass Properties	
Recuperator	23
Compressor	20
Turbine	9
Alternator	23
Conditioning	40
Radiator	236
Power conditioning	30
Total (mt)	381

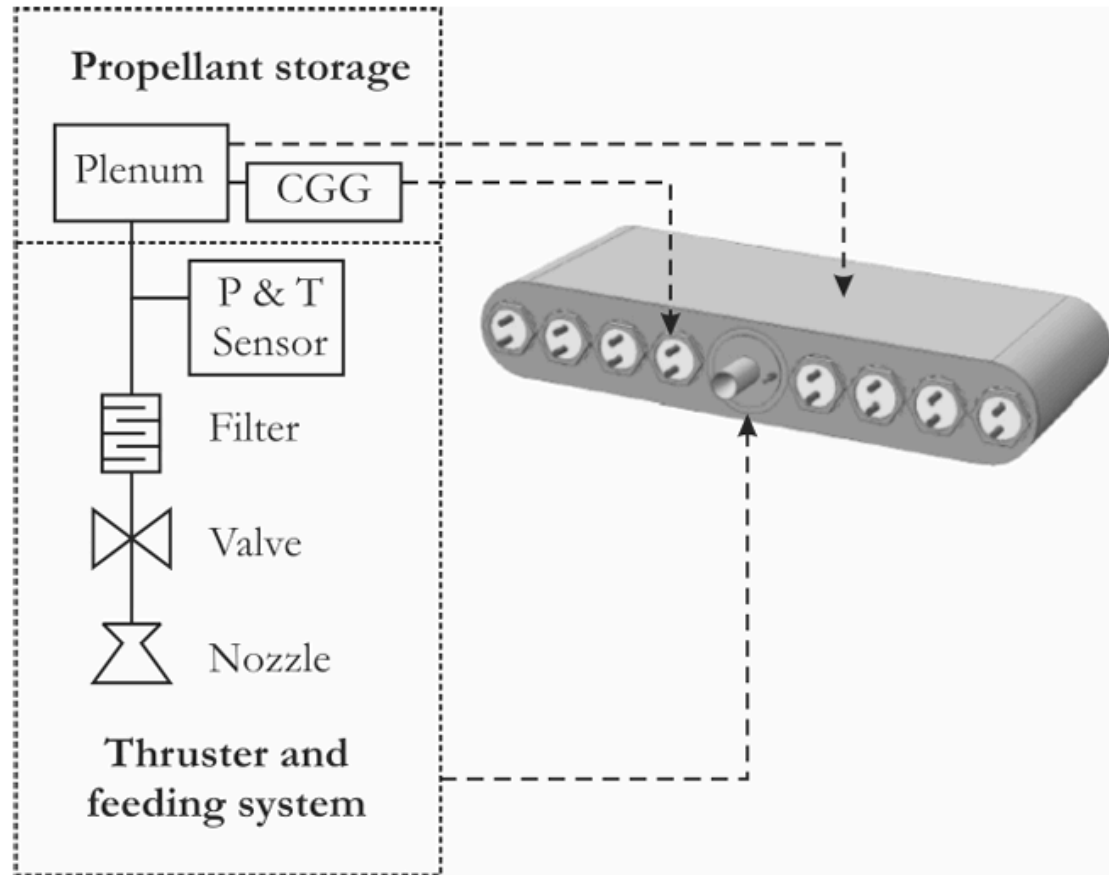
v_e is the velocity of the reaction mass when it leaves the satellite, m_i the initial mass and m_f the final mass. By rewriting equation (2.3), the total amount of reaction mass for the mission is obtained:

Variables: m m_{min} m_{max} Δv v_e $\Delta = - \cdot \cdot \cdot$ ifixit / e

Mission duration	1 year
Orbital altitude	500 km
Satellite mass / dimensions	1kg / 10x10x10 cm ³
Allowed drift in distance	2 m
Table 2-I: Mission characteristics	
Impulse bit	90 μ Ns
Number of corrections	6000
Total delta-v	0.54 m/s
Assumed exit velocity	673 m/s (Ma=4)
Reaction mass	0.8 g
Table 2-II: Propulsion requirements	

the functional requirements of a cold gas propulsion system are presented. Our micro propulsion system is a simple blow down system of a rocket engine consisting of a thruster and feeding system and a propellant storage tank. Besides the use of micro system technology to reduce the size of the thruster and feeding system, a novel technology has been developed which makes it possible to reduce the size as well

as the storage pressure of the propellant storage [5]. Cold gas generators (CGG) are used which contain a gas that is chemically stored inside a solid material. When ignited, nitrogen gas is released into a gas storage tank. Several CGG's are connected to the storage tank to refill the tank when the pressure drops below a proper operation pressure for the thruster.



Low storage pressure - which is enabled by the cool gas generator technology and is typically between 1 and 4.5 bar - makes a pressure regulator redundant. Such a regulator is normally required when using high-pressure tanks. Moreover, the low pressure makes storage easier and makes the design of the MST thruster and feeding system less critical in terms of fracture. To prevent system failure, if by accident more than one CGG is ignited and thus causing a high tank pressure, the system has to be able to withstand a pressure of 10bar.

1. Tank pressure Our propulsion system is a blowdown system meaning the pressure decreases during propulsion. The tank pressure is between 1 and 4.5 bar depending on the amount of gas in the tank. The pressure is maintained within this range by releasing nitrogen from a CGG when the operation pressure drops below 1 bar.

2. Gas temperature We are building a cold gas propulsion system without any thermal housekeeping. This means the gas temperature is mainly dependent on the temperature of the satellite and on the heat generated by the ignition of a CGG. When the satellite is orbiting the earth, it is for part of the time in eclipse and otherwise in direct sunlight. This causes periodic temperature changes of the satellite.

The tank pressure and temperature are measured to be able to determine the amount of thrust before the actual propulsion is performed. The impulse bit, which is the parameter that should be precisely controlled to be able to do formation flying.

When the satellite is in orbit two other environmental aspects are important namely, temperature and radiation. The operating range of the temperature depends on the thermal management of the satellite. In the extreme case the temperature can change 100°C from -40°C to 60°C in 90 minutes. The exposure to radiation is mainly important for sensitive electronic components which degrade over time due to radiation. But also mechanically, the protective body of the propulsion system should be thick enough to prevent puncture, and thus leakage, due to the impact of high-energy cosmic rays.

The amount of thrust that is produced depends on the mass flow rate $\dot{m}$, the exit velocity of the exhaust v_e and the pressure at the nozzle exit p_e . The thrust that is produced by a nozzle is calculated by [2]:

$$F = \dot{m} v_e + p_e A_e - p_a A_e$$

To determine the dimensions of the nozzle the following assumptions are made: The diverging part of the nozzle has a half angle (α) in between 15° and 20°.

A larger angle can cause rupture of the boundary layer and can result in shockwaves reducing the accuracy significantly. The converging part of the nozzle has a half angle of 30°.

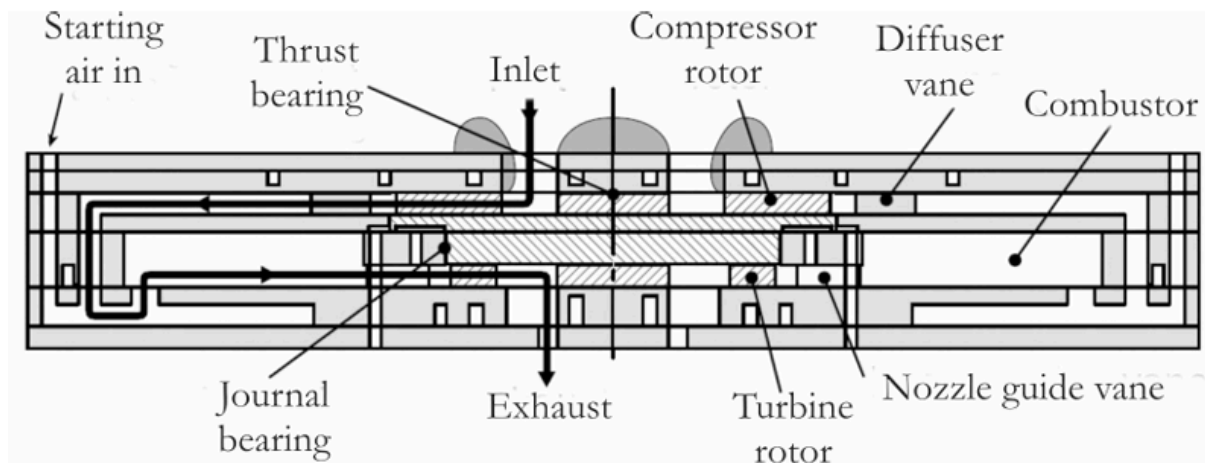
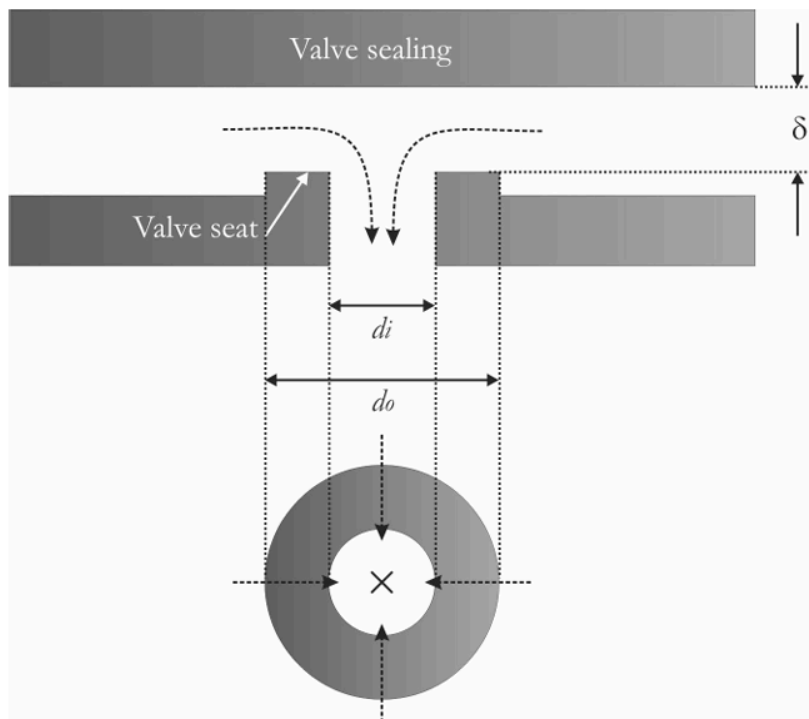
The converging part guides the gas towards the nozzle throat where it reaches Mach 1. Abrupt changes in the geometry should be avoided since it causes the thickness of the boundary layer to increase, and thereby reducing efficiency. MST is based on techniques to shape silicon and glass substrates.

Upper limit	Lower limit	
20° half angle	15° half angle	
Throat diameter (d_{throat})	131µm	90µm
Exit diameter	422µm	304µm
Thrust @ 4.5bar	10mN	4.72mN
Thrust @ 1bar	2.2mN	1.0mN
Expansion ratio	10.4	11.4
Mach-number	4	4.1

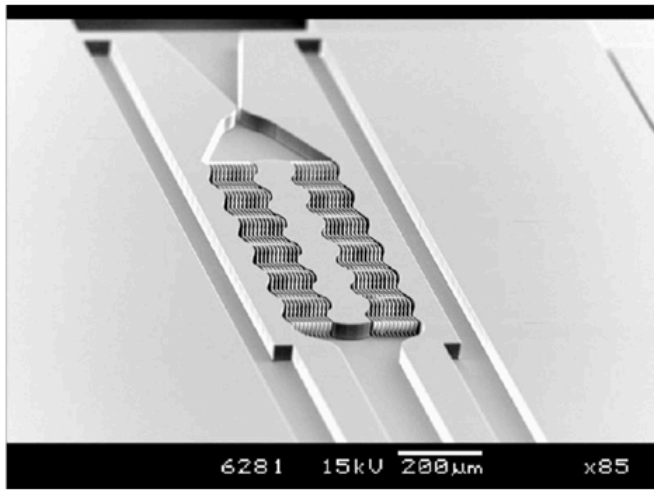
Valve An active normally closed valve is required to be able to accurately control the amount of propulsion. There are three main requirements that should be met:

1. The leakage of the valve in the closed state should be smaller than $1.6 \cdot 10^{-4}$ sccm as is earlier addressed. An in depth discussion about the leakage.
2. In the open state the area of passage should be at least 3 times higher than the nozzle throat area to avoid choked flow in the valve instead of in the nozzle.

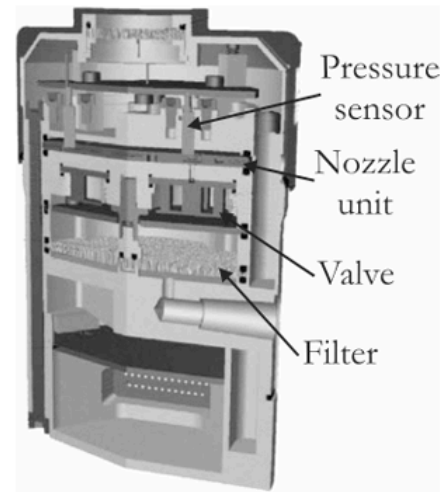
3. The valve should open and close within 1ms.



The baseline package of the modular platform is a glass tube bonded on silicon. Figure 3-3 shows the glass tube with a silicon component underneath it. This silicon component can contain any functionality according to the requirements of the system. In our case it is the valve module which is described in chapter 5. The glass tube is a convenient package because it is easily attached to silicon or other glass like materials by fusion bonding. It is functioning as a hermetically sealed package, fluidic interconnect as well as a macro support for the fragile MST components.



a)



b)

The glass tube is fixed on the MST part by fusion bonding at elevated temperature. First the glass tubes are diced to the right length and the bonding surface is polished. Just before fusion bonding the glass tube is ultrasonically cleaned in ethanol. The glass tube is positioned on top of the clean silicon valve. The heat treatment is done in a Nabertherm LH 15/12 furnace in an air environment. The unit is heated up to 790°C, maintained at this temperature for 30 minutes, after which it is gradually cooled down resulting in a hermetic seal between the glass tube and the silicon.